

SURVIVAL OF MID-LATITUDE GROUND ICE ON MARS. A. M. Bramson¹, S. Byrne¹, and J. Bapst¹,
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Introduction: In the last decade, many lines of evidence (Fig. 1) have emerged suggesting excess ice (higher water abundances than can fit into the pore spaces of the regolith) is common in the northern mid-latitudes (38-52°N) of Mars. Shallow Radar (SHARAD) sounding of the subsurface has detected dielectric interfaces that have been attributed to the bottom of excess ice sheets in Arcadia Planitia [1] and Utopia Planitia [2]. Geomorphological evidence includes thermokarstic features such as expanded craters [3] and scalloped depressions [4]. Many new impacts expose and excavate very pure water ice down to 38°N [5], and craters across Arcadia Planitia exhibit a terraced morphology due to layers in the subsurface that have been associated with the same ice layer detected by SHARAD [1]. Both Arcadia and Utopia deposits, each holding ice volumes on the order of 10s of thousands of km³, were found to have dielectric constants of ~2.5, consistent with excess ice and low dirt contents in the upper decameters of the surface.

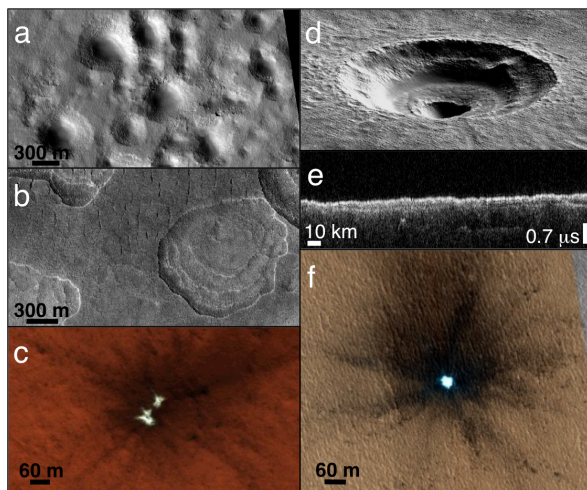


Figure 1. Evidence for excess ice in the northern mid-latitudes of Mars: (a) cluster of thermokarstically expanded craters; (b) scalloped depressions; (c & f) ice-exposing impacts; (d) 3D perspective of terraced crater indicating an impact into ice layers; (e) SHARAD radargram showing dielectric interfaces in the subsurface attributed to ice. HiRISE image credit: NASA/JPL/University of Arizona

The age of this ice in Arcadia Planitia was unexpectedly found to be at least 10s of Myr [3]. In this time, Mars has had many excursions through low obliquities, when low latitude ground ice gets transported to the poles. Martian regolith is predicted to

have high diffusion coefficients based on experimental studies [6, 7], suggesting ice should readily diffuse out of the subsurface during these periods of ice instability in the mid-latitudes. Replenishment of this ice from the atmosphere in subsequent epochs can only create pore-filling ice, not excess ice. Excess ice is thus expected to be geologically recent, and modeling of this process has found any ice sheets should be young and actively retreating towards today's conditions [8,9].

Understanding the conditions that have led to the excess ice's continued survival to the present day is important for understanding a) the stability of ice in the Amazonian (3 Ga to present), b) the subsurface structure in the mid-latitudes and c) the orbital forcing of the Martian climate. To test conditions for excess ice preservation in the last 10s of millions of years, we calculate the expected ice retreat throughout the 20 Myr-long orbital solutions (obliquity, eccentricity, and longitude of perihelion) of [10]. We will use our model to find the conditions necessary to reproduce the distribution, quantities and dielectric constants of excess ice deposits that are observed in the northern mid-latitudes.

Thermal Model: We model temperatures of the Martian subsurface using a 1D semi-implicit (e.g. Crank Nicholson) thermal conduction model that simulates surface energy balance and transfer of heat between subsurface layers. The model allows for subsurface layers of different thermophysical properties (e.g. conductivity, density, specific heat), which we use to model a surface porous regolith layer ("lag") atop excess ice.

From the temperatures of each numerical layer, we use the Clausius-Clapeyron equation and the ideal gas law to calculate the annual average vapor densities in the regolith at each orbital solution. We model atmospheric water vapor conditions following the obliquity-dependent scheme of [9]. When $\rho_{\text{vapor}} > \rho_{\text{atmos}}$ at the excess-ice table, ice is lost at a rate that is dependent on the vapor diffusivity of the regolith, the difference between the vapor density and atmospheric vapor content, and the depth of the ice table. When the excess ice retreats, it leaves behind any silicate material embedded within it, thus growing the lag deposit. We implement the growth of this regolith layer following the method described in [11].

Including the effect of the resupply of ice into pore-spaces within the lag deposit is important as it restricts further diffusion of the excess ice below. We allow vapor to fill the pore spaces of the regolith at and be-

low its equilibrium depth (where $\rho_{\text{vapor}} = \rho_{\text{atmos}}$) for cases when pore-filling ice is stable between the surface and the top of the excess ice layer. When pore-filling ice is stable, we assume no retreat of the excess ice sheet due to the pore-filling ice choking off exchange between the excess ice sheet and atmosphere.

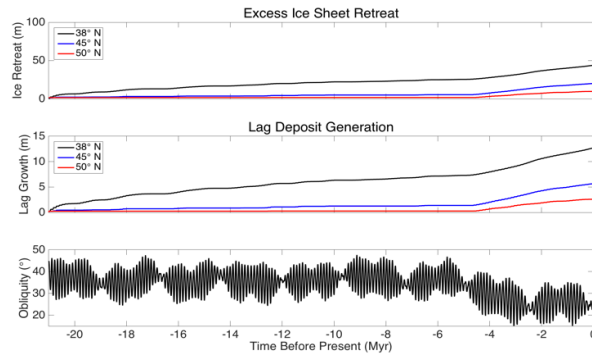


Figure 2. Ice retreat and lag deposit growth over time for excess ice purity of 85%.

Results: We calculate ice sheet retreat and lag deposit growth over 20 Myr for the latitude range of 38°–52°N and for different dirt contents (5–25%) within the excess ice layer. The dielectric constants of [1] and [2] require very low dirt contents. Low dirt content in the ice leads to a thinner lag deposit, which is favored by observational evidence from the ice-exposing impacts, as well as the low dielectric constants. However, lower dirt contents lead to higher amounts of ice retreat, making preservation of the ice sheet less likely.

For 45°N and 50°N, ice retreat is on the order of the amount of ice still present and lag thickness is a couple meters (Figure 2), suggesting preservation of the ice sheet over 10s of Myr is not unreasonable. Preserving the ice down to 38°N however, especially if covered by a thin lag deposit, is much more difficult. Some of the new impacts expose excess ice from depths of ~50 cm. Therefore, we also ran the model using a constant lag thickness (Figure 3), and find 600 m of ice loss if the lag deposit were held fixed at only 50 cm. To try to resolve this discrepancy in explaining ice preservation down to the lower mid-latitudes, we plan to test modifications in the atmospheric water vapor vs. obliquity scheme that may result in excess ice being more stable than previously thought.

Figure 4 shows the final ice retreats (top) and lag thicknesses (middle) across the latitude range of interest. We also calculate the dielectric constants for the final stratigraphies (bottom) using the dielectric mixing model of Stillman et al. 2010, which was used to determine the dielectric constant values in [1] and [2]. In this calculation, we assume the bottom of the excess

ice sheet today to be at 42 m, the average depth across Arcadia Planitia [1]. All of the simulated stratigraphies have dielectric constants higher than that measured across both Arcadia and Utopia. This suggests the ice sheet must have porosity (due to pore space dielectric constant being 1). Therefore, we are currently adding the effect of ice porosity to our model and will present these additional results.

The observations of newly-discovered mid-latitude ice deposits set current-day constraints on excess ice quantities and distributions. We will report on combinations of subsurface structure and Martian climate variability over the past 10s of Myr that correctly predict ice preservation.

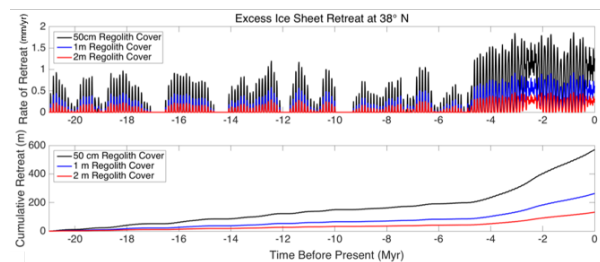


Figure 3. Ice retreat (rate and cumulative) for constant lag thicknesses (50 cm, 1 m and 3 m) at 38°N.

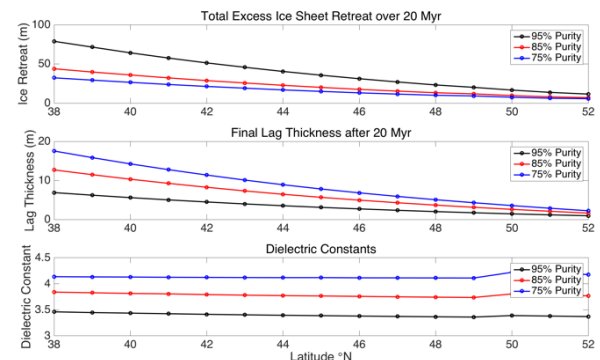


Figure 4. Final ice retreats (top), lag thicknesses (middle) and dielectric constants (bottom) across the latitude range for three purities of excess ice.

References: [1] Bramson A. M. et al. (2015) GRL, 42, 6566. [2] Stuurman C. M. et al. (2016) GRL, 43. [3] Viola D. et al. (2015) Icarus, 248, 190. [4] Dundas C. M. et al. (2015) Icarus, 262, 154. [5] Dundas C. M. et al. (2014) JGR-Planets, 119, 109. [6] Hudson T. L. et al. (2007) JGR, 112, E05016. [7] Bryson K. L. et al. (2008) Icarus, 196, 446. [8] Head J. W. et al. (2003) Nature, 426, 797–802. [9] Schorghofer N. and Forget F. (2012) Icarus, 220, 1112. [10] Laskar, J. et al. (2004) Icarus, 170, 343–364. [11] Schorghofer N. (2010) Icarus, 208, 598–607.